SUMMER TRAINING

Name: Dhanashri Shivdas   **College:** Indian Institute of Technology Delhi

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Project Statement: To Study transmission and distribution component and selection of switchgears, protection relays based on the line transmission & distribution losses, fault current calculation and relay coordination from upstream to downstream as outcome.

Introduction:

An efficient and reliable electrical power system hinges on the optimal design and coordination of transmission and distribution (T&D) components, with increasing power demands and greater network complexity, the use of properly selected switchgear and precisely coordinated protection relays become essential. This research undertaken at the Energy Plant seeks to enhance the reliability and performance of the plant's T & D system by minimizing power losses, selecting appropriate equipment, and improving protection through relay coordination. The analysis spans the complete T & D network, extending from the generator terminals to load feeders. Core Activities involve study of components of T&D of the system, Trip Time calculation of breakers, power loss in transmission and distribution of the overall plant and fault current calculation. These form the basis for selection and configuration of Circuit Breakers, Isolators, Current Transformers and Protection Relays, ensuring comprehensive safety and selection isolation. This research blends theoretical grounding with practical evaluation, yielding valuable insights for the Energy Plant and other industrial electrical networks.

Objectives:

- Catalog and assess all T&D components at the plant.
- Conduct detailed fault current analysis using site data.
- Optimize protective relay settings from the source to end-user points.
- Choose appropriate switchgear based on load and fault analysis.
- Evaluate losses in T&D Network and improve efficiency of the network.
- Recommend reliability-enhancing interventions.

Methodology:

A structured methodology integrates field surveys, theoretical computations, and software simulations.

1. Data Collection: Gather Single Line Diagrams (SLDs), relay settings, transformer specifications, and generator ratings from the plant.
2. System Modelling: Develop a comprehensive system model, segmented into logical subparts for simplified analysis and calculations.
3. Fault Analysis: Apply per-unit and symmetrical components techniques to calculate fault current throughout the network.
4. Trip Time Calculation: Calculate trip time based on existing relay settings and evaluate the relay coordination performance.
5. Relay Settings Optimization: Optimize pickup settings, Time Multiplier Settings (TMS), and Coordination Time Intervals (CTI) values based on breaker trip sequences from upstream to downstream.
6. Switchgear Evaluation: Analyze the operational capacity of existing circuit breakers and transformers; reconfiguration to enhance system reliability and performance.
7. Review and Validation: Compare outcomes with actual performance data and propose effective system improvements based on findings.

System description:

The electrical infrastructure at Energy Plant includes:

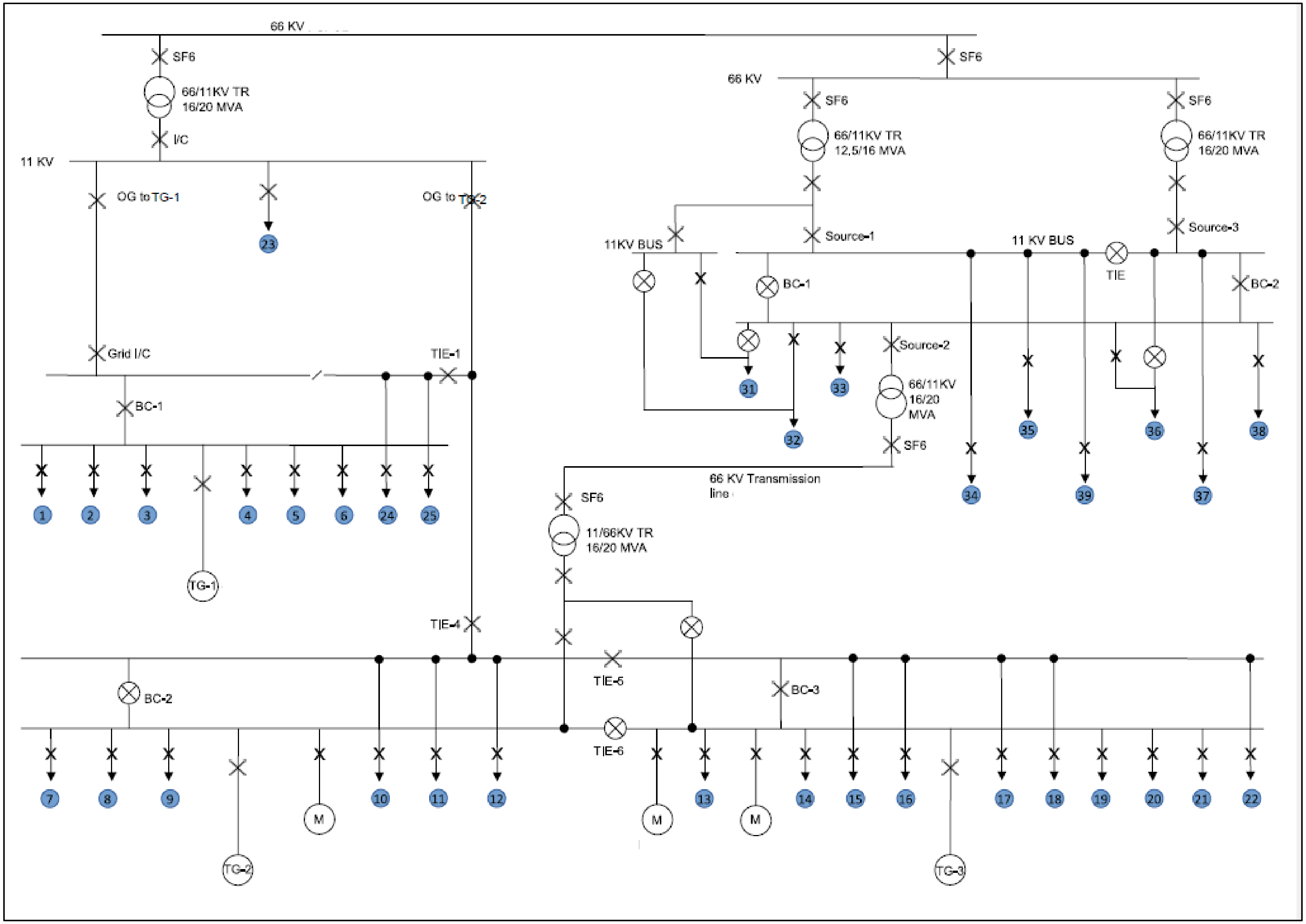
- Generators: 2 Generators are 20 MW, and one generator is 11.3 MW.
- Transmisson line: 66kV transmission line of 20 MW capacity having length of 13km.
- Transformers: Step-up and Step-down Transformers for grid interface, auxiliary Transformers for internal distribution.
- Busbars: Main and Auxiliary configurations for redundancy.
- Feeders: Distribution lines to Different plant loads.
- Switchgear: ACBs, VCBs, and Numerical Relays (MICOM P111, P122, P127, P422) (SPAJ 130C,140C,150C), (SIEMENS 7UM622)
- Loads: Include motors, lightning, and HVAC systems.

Each Zone is monitored by CTs and PTs to detect abnormal conditions and relay them to protection devices.

Description of SLD of Overall System:

The SLD shown above has 2 generators of 20 MW along with 1 generator of 11.3 MW arranged in parallel. Each generator feeds its own 11 KV bus section through a dedicated breaker (BC1, BC2, BC-3) and protection relay. From each generator we can track:

- **Grid Interconnection –** A tie breaker to the utility grid. This allows import of power and provides an alternate source if a generator trips.
- **Transformer Feeders** – Outgoing lines to step-down transformers that supply auxiliary, and process loads at lower voltage.
- **Feeder Panels** – Arrays of feeder breakers supplying major plant loads. Each feeder has its own overcurrent and earth-fault relay.
- **Spare and Tie-Over Feeders** – Spare bays and occasional inter-tie feeders for flexibility during maintenance or fault isolation.



Information of Relay Placed on Different Feeders:

- ## Types of feeders:

1. Source Feeder:

- A feeder that connects the **power source** to the distribution system. The purpose of this feeder is to carry bulk power from the main generation unit (like Generator) to other components such as transformers or main busbars.
- These Feeders have **high breaking capacity** and sensitive protection to detect faults immediately at the source, in which relays like **SIPROTEC 7UM62** are commonly used.
- In this study, **Gen I/C, Grid I/C, Bus Coupler and Tie Fitter** all these feeders come under this category.

2. Motor Feeder:

- Supplies power directly to **large Motors or motor control centers**; the purpose of these feeders is to deliver power to process equipment such as pumps, drives, compressors, etc.
- Relays Placed on these feeders must protect against **overload, locked rotor, and phase imbalance**. Also, relay settings are tuned such that motor allows starting inrush currents while still providing fault protection.
- O/G feeder drives are this kind of feeder in current study.

3. O/G Feeder:

- Distributes power to external loads or different sections of the plant; acts as the output arm from the main switchgear panel or bus to load centers like O/G feeders.
- They require precise relay coordination to ensure selectivity so that faults in one section do not trip upstream breakers unnecessarily. For this purpose, relays like **MICOM P111/P122** are used.

4. Transformer Feeder:

- Connect switchgear to power transformers which step voltage up or down. Feed power to transformers (e.g. **5MVA T/F, 2.5 MVA T/F**) to distribute at different voltage levels for various plant operations.
- Critical for protecting transformers from **short circuits, inrush currents, overloading, and differential faults**. CT ratios and relay curves must be considered Transformer characteristics.

- ## Types of Relays:

1. Numerical Relays:

- These are **Microprocessor based devices** that provide multifunction protection with high accuracy. They support communication protocols such as **Modbus and IEC 61850**, making them suitable for integration into **SCADA** systems.
- These relays are programmable, allowing flexible configuration for different protection needs. At Energy Plant, they are mainly used for feeders, transformers, and motor protection due to their versality, reliability, and ease of monitoring.
- **Micom p111 and p122** come under this category.

2. SPAJ Relays (ABB Series):

- These are earlier generation **digital/numerical protection relays** designed primarily for overcurrent and earth fault protection.

- They are known for their **compact design, rugged build, and user-friendly LCD interface**, making them suitable for harsh industrial environments.
- In the plant, SPAJ relays are typically installed on **feeder panels and auxiliary load circuits** where reliable and straightforward protection is essential.
- Their durability and simplicity make them ideal for legacy systems still in operation. This includes **SPAJ 130C, SPAJ 140C, and SPAJ 150C**.

3. Electromechanical Relays:

- These relays are traditional protection devices that operate on the **principle of electromagnetic induction**. They are simple in design and known for their mechanical reliability, but they are **relatively slow, bulky, and lack modern communication features**.
- Due to their l**imited configurability and the need for manual testing**, they are not preferred in modern systems.
- In the plant, these relays are mostly found in older installations or used as backup protection in non-critical areas where **basic fault protection** is sufficient.

4. Generator Protection Relays:

- These relays are generator protection relays which are **advanced digital relays** specifically designed for safeguarding synchronous generators.
- They provide comprehensive protection functions including **overcurrent, differential protection, loss of excitation, unbalanced load, and reverse power**, ensuring reliability and system stability.
- With **high-speed tripping, event logging, and compatibility with modern communication protocols (like IEC 61850)**, these relays are essential for critical applications.
- In the plant, such relays are used at generator output terminals to ensure secure and responsive protection from the very point of power generation.

Information of Relays Placed on Different Feeders in the plant:

GENERATOR-3

Feeder Name	Relay Type	CT ratio
GENERATOR	SIPROTEC 7UM62	1600/1
BUS COUPLER	MICOM P111	1600/1
TR1	MICOM P111	600/1
TR2	MICOM P111	300/1
O/G Feeder 22	MICOM P111	75/1
O/G Feeder 20	MICOM P122	600/1
O/G Feeder 18	MICOM P111	600/1

O/G Feeder 17	MICOM P111	600/1
O/G Feeder 16	MICOM P111	1200/1
O/G Feeder 15	MICOM P111	1200/1
O/G Feeder 14	MICOM P111	1200/1
O/G Feeder 13	MICOM P122	1600/1
TIE-5	MICOM P127	1600/1
TIE-4	SPAJ 140C ABB	1000/1
GRID TG 2/3	MICOM P127	Line-1200/1
		Ground-50/1
11KV GRID I/C	MICOM P127	Line-1200/1
		Ground-50/1
66 KV RELAY SF6	MICOM P127	200/1

GENERATOR-1

Feeder Name	Relay Type	CT ratio
GENERATOR	SIPROTEC 7UM62	1600/1
BUS COUPLER	Electromechanical relay	
TR1	Electromechanical relay	
TR2	MICOM P111	150/1
O/G feeder 24	Display not working	
O/G Feeder 5	SPAJ 140C ABB	400/1
O/G Feeder 4	MICOM P111	400/1
O/G Feeder 3	MICOM P111	300/1
GRID I/C	MICOM P127	600/1
GRID TG-1	MICOM P127	1200/1
11KV GRID I/C	MICOM P127	1200/1
66 KV RELAY SF6	MICOM P127	200/1

GENERATOR-2

Feeder Name	Relay Type	CT ratio
GENERATOR	SIPROTEC 7UM62	1000/1
BUS COUPLER		unknown
TR1	SPAJ 140 C	600/1
TR2	SPAJ 140 C	600/1
O/G Feeder 12	MICOM P111	600/1
O/G Feeder 11	MICOM P122	600/1
O/G Feeder 10	MICOM P111	600/1
O/G Feeder 9	MICOM P122	800/1
Power Outlet	MICOM P127	1200/1
11KV TRANSFORMER	MICOM P122	1200/1
66 KV SF-6 Breaker	MICOM P122	200/1
66 KV SF-6 Breaker	MICOM P122	200/1

1. Feed Pump (BFP-1, BFP-2, BFP-3):

- **No Overcurrent Pickup (I>) timing Available**: The relay configuration for the boiler feed does not include a definite I> delay for inverse time operation.
- These relays are often configured **only with instantaneous overcurrent (I>>) protection**, as feed pumps are highly critical equipment.
- Therefore, their **trip time is governed only by I>>**, and this **instantaneous operation is not suitable** for inclusion in coordinated time grading, which relies on **graded I> delays** across the system.

2. Main generator incomer feeders (Main Gen I/C TG-1, TG-2, TG-3):

- These **main generator incomer** relays are localized protection that act **only within the generator section** and don't participate in downstream protection grading.
- The generators are **isolated protection zones** (islanded when needed), and their relay operations are not supposed to influence or be influenced by **downstream trip coordination**.
- Including them in the grading sequence could **distort relay selectivity** across the plant, as they protect **generation equipment**, not transmission or distribution feeders.

Relay Settings of Various Parameters:

1. **Overcurrent Pickup Value (I>):** The minimum current (in per unit or multiple of rated current) at which the relay starts to operate.
2. **Time Multiplier Settings (TMS):** A scaling factor used in the IDMT (Inverse Definite Minimum Time) curve equation to adjust operating time. Higher TMS will result in longer Trip Time.
3. **Operating Time for Overcurrent Pickup (Time I>):** Actual time (in seconds) at which the relay tripping once the current exceeds the pickup value, unless the condition is cleared.
4. **Instantaneous Overcurrent Pickup (I>>):** A higher preset current level at which the relay trips without the intentional delay to respond rapidly to severe faults.
5. **Trip Time for Instantaneous Overcurrent (Time I>>):** Instantaneous or definite trip delay for high-set overcurrent. Usually, it is very small.
6. **Earth Fault Pickup (Ie>):** Sensitivity level pf relay to detect earth faults; a lower value provides greater sensitivity to ground leakage current.
7. **Trip Time for Earth Fault Pickup (Time Ie>):** The delay time before the relay operates under an earth fault condition at the Ie> pickup level.
8. **Instantaneous Earth Fault Pickup (Ie>>):** Higher settings for detecting severe earth fault, enabling immediate relay response without time delay.
9. **Trip Time for Instantaneous Earth Fault Pickup (Time Ie>>):** The associated trip Delay for the Ie>> threshold, allowing rapid isolation during high intensity earth faults.
10. **Current Transformer Ratio (CT):** The ratio used to convert actual primary current to a manageable secondary current level for relay metering and protection.

GENERATOR – 3

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~
BUS COUPLER	MICOM P111	1600/1	0.8	0.25	0.8	0.25	1.7	0.4	0.06	0.12
TR1	MICOM P111	600/1	0.4	0.5	2.5	0.04	0.06	0.04	0.1	0.02
TR2	MICOM P111	300/1	0.4	0.04	0.7	0.02	0.1	0.04	0.04	0.1
O/G Feeder 22	MICOM P111	75/1	1.2	0.04	4	0.04	0.2	0.04	0.25	0.02
O/G Feeder 20	MICOM P122	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02
O/G Feeder 18	MICOM P111	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02
O/G Feeder 17	MICOM P111	600/1	0.7	0.04	1	0.002	0.06	0.04	0.1	0.02
O/G Feeder 16	MICOM P111	1200/1								
O/G Feeder 15	MICOM P111	1200/1	0.5	0.04	1.2	0.02	0.06	0.04	0.06	0.02

O/G Feeder 14	MICOM P111	1200/1	0.4	0.04	1.7	0.02	0.03	0.04	0.06	0.02
O/G Feeder 13	MICOM P122	1600/1	0.15	0.025	0.5	0.05	0.025	0.025	0.025	0.02
TIE-5	MICOM P127	1600/1	0.7	0.04	2.1	0.04	0.06	0.12	0.07	0.35
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35
GRID TG 2/3	MICOM P127	Line - 1200/1 Ground- 50/1	0.84	0.14	3.33	0.45	0.85	0.12	0.85	0.35
11KV GRID I/C	MICOM P127	Line - 1200/1 Ground- 50/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2

GENERATOR – 1

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~
BUS COUPLER	Electromechanical relay									
TR1	Electromechanical relay									
TR2	MICOM P111	150/1	0.7	0.07	4	0.02	0.15	0.05	0.2	0.05
O/G feeder 24	Display not working									
O/G Feeder 5	SPAJ 140C ABB	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.05
O/G Feeder 4	MICOM P111	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.04
O/G Feeder 3	MICOM P111	300/1	1.25	0..05	2	0.05	0.12	0.05	0.12	0.04
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1
GRID TG-1	MICOM P127	1200/1	0.8	0.14	3	0.3	0.1	0.12	0.1	0.15
11KV GRID I/C	MICOM P127	1200/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2

GENERATOR – 2

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)
GENERATOR	SIPROTEC 7UM62	1000/1	0.92	0.8	1.31	0.1	0.33	0.2	~	~
TR1	SPAJ 140 C	600/1	0.54	0.13	4	0.04	0.15	0.12	0.17	0.1
TR2	SPAJ 140 C	600/1	0.5	0.13	4	0.04	0.15	0.12	0.17	0.1
O/G Feeder 12	MICOM P111	600/1	0.7	0.05	1.2	0.05	0.1	0.02	0.12	0.02
O/G Feeder 11	MICOM P122	600/1	0.4	0.04	0.5	0.05	0.05	0.025	0.05	0.02
O/G Feeder 10	MICOM P111	600/1	0.5	0.05	0.5	0.04	0.1	0.05	0.1	0.05
O/G Feeder 9	MICOM P122	800/1	1	0.04	1.8	0.04	0.1	0.025	0.05	0.02
Power Outlet	MICOM P127	1200/1	0.9	0.07	1.8	0.08	0.06	0.04	0.1	0.05
11KV TRANSFORMER	MICOM P122	1200/1	0.9	0.07	1.7	0.07	0.1	0.05	0.1	0.05
66 KV SF-6 Breaker	MICOM P122	200/1	0.9	0.05	1.6	0.08	0.1	0.07	0.1	0.07

Key Relay Coordination Principles:

- **Time Grading** – Each downstream relay has a slightly shorter trip time at a given fault current than its upstream neighbor.
- **Current Grading** – Pickup settings (I>, I>>, earth fault) increases as you move upstream so that small faults are handled close to the load.
- **CTI Margin** – A minimum margin of 0.3s is maintained between stages to account for relay operating time tolerances and breaker opening times.
- **Selective isolation** – Only the nearest breaker to a fault operates, minimizing service interruption elsewhere in the plant.

Sequence of Tripping Time for each generator:

Name of the Generator	Sequence of the Feeders	Direction based on trip time
GEN – 3	• TG-3 I/C • Bus Coupler – 3 • O/G feeders – 1 • O/G feeders – 2 • O/G feeders – 3 • Tie • Grid I/C	↑
GEN – 3	• Grid I/C • Grid power to TG-3 • 11KV Grid I/C	↓
GEN – 3	• 11KV Grid I/C • 66 KV relay SF6	↑
GEN – 1	• TG-1 I/C • Bus Coupler – 1 • O/G feeders – 1 • O/G feeders – 2 • O/G feeders – 3 • TR-1 • TR-2 • Grid I/C	↑
GEN – 1	• Grid I/C Grid power to TG-1 • 11KV Grid I/C	↓
GEN – 1	• 11KV Grid I/C • 66 KV relay SF6	↑

GEN – 2	• TG-3 I/C • Bus Coupler – 2 • O/G feeders – 1 • O/G feeders – 2 • O/G feeders – 3 • Tie • Power O/G	
GEN – 2	• 11KV Transformer O/G breaker • 66 KV relay SF6	
GEN – 2	• 66 KV relay SF-6 • 11 KV Transformer O/G breaker • 11 KV I/C - DHC Power • 11 KV I/C -Discom Power • 66 KV SF-6 Discom Power	

Trip Time Calculation of Overcurrent Using IEC 60255 Standard Inverse Curve:

Relay coordination ensures timely and selective fault clearance. Key parameters:

1. Pickup Current = Value of Overcurrent × secondary parameter value of CT ratio.
2. Trip Time can be calculated by below formula:

$$Trip\ Time\ = TMS \times \frac{0.14}{\left(\left(\frac{Fault\ Current}{Pickup\ \ Current}\right)^{0.02} - 1\right)}$$

3. Maintain a CRI of 0.3-0.5 s between relays.
4. CTs are rated based on maximum expected current.
5. For the calculations Fault current is taken as 5000 Amp.

Trip Time Calculation for Overcurrent:

GENERATOR –3

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	5.898
BUS COUPLER	MICOM P111	1600/1	0.8	0.25	0.8	0.25	1.7	0.4	0.06	0.12	1.267
TR1	MICOM P111	600/1	0.4	0.5	2.5	0.04	0.06	0.04	0.1	0.02	1.118
TR2	MICOM P111	300/1	0.4	0.04	0.7	0.02	0.1	0.04	0.04	0.1	0.072
O/G Feeder 22	MICOM P111	75/1	1.2	0.04	4	0.04	0.2	0.04	0.25	0.02	0.067
O/G Feeder 20	MICOM P122	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02	0.104
O/G Feeder 18	MICOM P111	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02	0.104
O/G Feeder 17	MICOM P111	600/1	0.7	0.04	1	0.002	0.06	0.04	0.1	0.02	0.11
O/G Feeder 16	MICOM P111	1200/1	0.5	0.04	1.2	0.02	0.06	0.04	0.06	0.02	0.129
O/G Feeder 15	MICOM P111	1200/1	0.4	0.04	1.7	0.02	0.03	0.04	0.06	0.02	0.117
O/G Feeder 14	MICOM P122	1600/1	0.15	0.025	0.5	0.05	0.025	0.025	0.025	0.02	0.056
TIE-5	MICOM P127	1600/1	0.7	0.04	2.1	0.04	0.06	0.12	0.07	0.35	0.184
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35	0.514

**Main Grid Incomer(tie-4) has a decoupling relay connected with it, which decouples it from rest of the system when needed hence it has greater trip time, it provides backup protection.*

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35	0.514
GRID TG 2/3	MICOM P127	Line - 1200/1 Ground- 50/1	0.84	0.14	3.33	0.45	0.85	0.12	0.85	0.35	0.602
11KV GRID I/C	MICOM P127	Line - 1200/1 Ground- 50/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35	0.7

66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2	0.398

GENERATOR –1

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	3.252
TR2	MICOM P111	150/1	0.7	0.07	4	0.02	0.15	0.05	0.2	0.05	0.122
O/G Feeder 5	SPAJ 140C ABB	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.05	0.135
O/G Feeder 4	MICOM P111	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.04	0.135
O/G Feeder 3	MICOM P111	300/1	1.25	0..05	2	0.05	0.12	0.05	0.12	0.04	0.132
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1	0.484

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1	0.484
GRID TG-1	MICOM P127	1200/1	0.8	0.14	3	0.3	0.1	0.12	0.1	0.15	0.584
11KV GRID I/C	MICOM P127	1200/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35	0.7
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2	0.398

GENERATOR –2

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1000/1	0.92	0.8	1.31	0.1	0.33	0.2	~	~	5.898
TR1	SPAJ 140 C	600/1	0.54	0.13	4	0.04	0.15	0.12	0.17	0.1	0.42
TR2	SPAJ 140 C	600/1	0.5	0.13	4	0.04	0.15	0.12	0.17	0.1	0.314
O/G Feeder 12	MICOM P111	600/1	0.7	0.05	1.2	0.05	0.1	0.02	0.12	0.02	0.138
O/G Feeder 11	MICOM P122	600/1	0.4	0.04	0.5	0.05	0.05	0.025	0.05	0.02	0.089
O/G Feeder 10	MICOM P111	600/1	0.5	0.05	0.5	0.04	0.1	0.05	0.1	0.05	0.121
O/G Feeder 9	MICOM P122	800/1	1	0.04	1.8	0.04	0.1	0.025	0.05	0.02	0.15
Power Outlet	MICOM P127	1200/1	0.9	0.07	1.8	0.08	0.06	0.04	0.1	0.05	0.315

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
11KV TRANSFORMER	MICOM P122	1200/1	0.9	0.07	1.7	0.07	0.1	0.05	0.1	0.05	0.315
66 KV SF-6 Breaker	MICOM P122	200/1	0.9	0.05	1.6	0.08	0.1	0.07	0.1	0.07	0.065

Trip Time Calculation for Instantaneous Overcurrent:

GENERATOR -3

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	0.1
BUS COUPLER	MICOM P111	1600/1	0.8	0.25	0.8	0.25	1.7	0.4	0.06	0.12	0.25
TR1	MICOM P111	600/1	0.4	0.5	2.5	0.04	0.06	0.04	0.1	0.02	0.04
TR2	MICOM P111	300/1	0.4	0.04	0.7	0.02	0.1	0.04	0.04	0.1	0.02
O/G Feeder 22	MICOM P111	75/1	1.2	0.04	4	0.04	0.2	0.04	0.25	0.02	0.04
O/G Feeder 20	MICOM P122	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02	0.02
O/G Feeder 18	MICOM P111	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02	0.02
O/G Feeder 17	MICOM P111	600/1	0.7	0.04	1	0.002	0.06	0.04	0.1	0.02	0.002
O/G Feeder 16	MICOM P111	1200/1	0.5	0.04	1.2	0.02	0.06	0.04	0.06	0.02	0.02
O/G Feeder 15	MICOM P111	1200/1	0.4	0.04	1.7	0.02	0.03	0.04	0.06	0.02	0.02
O/G Feeder 14	MICOM P122	1600/1	0.15	0.025	0.5	0.05	0.025	0.025	0.025	0.02	0.05
TIE-5	MICOM P127	1600/1	0.7	0.04	2.1	0.04	0.06	0.12	0.07	0.35	0.04
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35	0.4

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35	0.4
GRID TG 2/3	MICOM P127	Line - 1200/1 Ground- 50/1	0.84	0.14	3.33	0.45	0.85	0.12	0.85	0.35	0.45

11KV GRID I/C	MICOM P127	Line - 1200/1 Ground-50/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35	0.5
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2	0.1

GENERATOR –1

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	0.1
TR2	MICOM P111	150/1	0.7	0.07	4	0.02	0.15	0.05	0.2	0.05	0.02
O/G Feeder 5	SPAJ 140C ABB	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.05	0.04
O/G Feeder 4	MICOM P111	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.04	0.04
O/G Feeder 3	MICOM P111	300/1	1.25	0..05	2	0.05	0.12	0.05	0.12	0.04	0.05
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1	1

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1	1
GRID TG-1	MICOM P127	1200/1	0.8	0.14	3	0.3	0.1	0.12	0.1	0.15	0.3
11KV GRID I/C	MICOM P127	1200/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35	0.5
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2	0.1

GENERATOR –2

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1000/1	0.92	0.8	1.31	0.1	0.33	0.2	~	~	0.1
TR1	SPAJ 140 C	600/1	0.54	0.13	4	0.04	0.15	0.12	0.17	0.1	0.04
TR2	SPAJ 140 C	600/1	0.5	0.13	4	0.04	0.15	0.12	0.17	0.1	0.04
O/G Feeder 12	MICOM P111	600/1	0.7	0.05	1.2	0.05	0.1	0.02	0.12	0.02	0.05

O/G Feeder 11	MICOM P122	600/1	0.4	0.04	0.5	0.05	0.05	0.025	0.05	0.02	0.05
O/G Feeder 10	MICOM P111	600/1	0.5	0.05	0.5	0.04	0.1	0.05	0.1	0.05	0.04
O/G Feeder 9	MICOM P122	800/1	1	0.04	1.8	0.04	0.1	0.025	0.05	0.02	0.04
Power Outlet	MICOM P127	1200/1	0.9	0.07	1.8	0.08	0.06	0.04	0.1	0.05	0.08

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
11KV TRANSFORMER	MICOM P122	1200/1	0.9	0.07	1.7	0.07	0.1	0.05	0.1	0.05	0.07
66 KV SF-6 Breaker	MICOM P122	200/1	0.9	0.05	1.6	0.08	0.1	0.07	0.1	0.07	0.08
66 KV SF-6 Breaker	MICOM P122	200/1	0.9	0.05	1.6	0.08	0.1	0.07	0.1	0.07	0.08

Trip Time Calculation for Earth Current:

GENERATOR –3

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	0.7
BUS COUPLER	MICOM P111	1600/1	0.8	0.25	0.8	0.25	1.7	0.4	0.06	0.12	4.571
TR1	MICOM P111	600/1	0.4	0.5	2.5	0.04	0.06	0.04	0.1	0.02	0.054
TR2	MICOM P111	300/1	0.4	0.04	0.7	0.02	0.1	0.04	0.04	0.1	0.052
O/G Feeder 22	MICOM P111	75/1	1.2	0.04	4	0.04	0.2	0.04	0.25	0.02	0.045
O/G Feeder 20	MICOM P122	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02	0.054
O/G Feeder 18	MICOM P111	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02	0.054
O/G Feeder 17	MICOM P111	600/1	0.7	0.04	1	0.002	0.06	0.04	0.1	0.02	0.054
O/G Feeder 16	MICOM P111	1200/1	0.5	0.04	1.2	0.02	0.06	0.04	0.06	0.02	0.063
O/G Feeder 15	MICOM P111	1200/1	0.4	0.04	1.7	0.02	0.03	0.04	0.06	0.02	0.054
O/G Feeder 14	MICOM P122	1600/1	0.15	0.025	0.5	0.05	0.025	0.025	0.025	0.02	0.035
TIE-5	MICOM P127	1600/1	0.7	0.04	2.1	0.04	0.06	0.12	0.07	0.35	0.204
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35	0.231

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35	0.231
GRID TG 2/3	MICOM P127	Line - 1200/1 Ground- 50/1	0.84	0.14	3.33	0.45	0.85	0.12	0.85	0.35	0.168
11KV GRID I/C	MICOM P127	Line - 1200/1 Ground- 50/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35	0.203
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2	0.207

GENERATOR –1

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	0.7
TR2	MICOM P111	150/1	0.7	0.07	4	0.02	0.15	0.05	0.2	0.05	0.061
O/G Feeder 5	SPAJ 140C ABB	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.05	0.069
O/G Feeder 4	MICOM P111	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.04	0.069
O/G Feeder 3	MICOM P111	300/1	1.25	0..05	2	0.05	0.12	0.05	0.12	0.04	0.067
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1	0.218

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1	0.218
GRID TG-1	MICOM P127	1200/1	0.8	0.14	3	0.3	0.1	0.12	0.1	0.15	0.217
11KV GRID I/C	MICOM P127	1200/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35	0.677
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2	0.258

GENERATOR –2

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	0.7
TR1	SPAJ 140 C	600/1	0.54	0.13	4	0.04	0.15	0.12	0.17	0.1	0.201
TR2	SPAJ 140 C	600/1	0.5	0.13	4	0.04	0.15	0.12	0.17	0.1	0.201
O/G Feeder 12	MICOM P111	600/1	0.7	0.05	1.2	0.05	0.1	0.02	0.12	0.02	0.03
O/G Feeder 11	MICOM P122	600/1	0.4	0.04	0.5	0.05	0.05	0.025	0.05	0.02	0.032
O/G Feeder 10	MICOM P111	600/1	0.5	0.05	0.5	0.04	0.1	0.05	0.1	0.05	0.076
O/G Feeder 9	MICOM P122	800/1	1	0.04	1.8	0.04	0.1	0.025	0.05	0.02	0.041
Power Outlet	MICOM P127	1200/1	0.9	0.07	1.8	0.08	0.06	0.04	0.1	0.05	0.063

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
11KV TRANSFORMER	MICOM P122	1200/1	0.9	0.07	1.7	0.07	0.1	0.05	0.1	0.05	0.09
66 KV SF-6 Breaker	MICOM P122	200/1	0.9	0.05	1.6	0.08	0.1	0.07	0.1	0.07	0.084
66 KV SF-6 Breaker	MICOM P122	200/1	0.9	0.05	1.6	0.08	0.1	0.07	0.1	0.07	0.084

Trip Time Calculation for Instantaneous Earth Current:

GENERATOR –3

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	~
BUS COUPLER	MICOM P111	1600/1	0.8	0.25	0.8	0.25	1.7	0.4	0.06	0.12	0.12
TR1	MICOM P111	600/1	0.4	0.5	2.5	0.04	0.06	0.04	0.1	0.02	0.02
TR2	MICOM P111	300/1	0.4	0.04	0.7	0.02	0.1	0.04	0.04	0.1	0.1
O/G Feeder 22	MICOM P111	75/1	1.2	0.04	4	0.04	0.2	0.04	0.25	0.02	0.02
O/G Feeder 20	MICOM P122	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02	0.02
O/G Feeder 18	MICOM P111	600/1	0.6	0.04	1	0.02	0.06	0.04	0.1	0.02	0.02
O/G Feeder 17	MICOM P111	600/1	0.7	0.04	1	0.002	0.06	0.04	0.1	0.02	0.02

O/G Feeder 16	MICOM P111	1200/1	0.5	0.04	1.2	0.02	0.06	0.04	0.06	0.02	0.02
O/G Feeder 15	MICOM P111	1200/1	0.4	0.04	1.7	0.02	0.03	0.04	0.06	0.02	0.02
O/G Feeder 14	MICOM P122	1600/1	0.15	0.025	0.5	0.05	0.025	0.025	0.025	0.02	0.02
TIE-5	MICOM P127	1600/1	0.7	0.04	2.1	0.04	0.06	0.12	0.07	0.35	0.35
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35	0.35

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
TIE-4	SPAJ 140C ABB	1000/1	1	0.12	3.9	0.4	0.15	0.12	0.12	0.35	0.35
GRID TG 2/3	MICOM P127	Line - 1200/1 Ground- 50/1	0.84	0.14	3.33	0.45	0.85	0.12	0.85	0.35	0.35
11KV GRID I/C	MICOM P127	Line - 1200/1 Ground- 50/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35	0.35
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2	0.2

GENERATOR –1

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1600/1	1.22	0.8	1.75	0.1	0.44	0.2	~	~	~
TR2	MICOM P111	150/1	0.7	0.07	4	0.02	0.15	0.05	0.2	0.05	0.05
O/G Feeder 5	SPAJ 140C ABB	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.05	0.05
O/G Feeder 4	MICOM P111	400/1	1	0.05	4	0.04	0.1	0.05	0.12	0.04	0.04
O/G Feeder 3	MICOM P111	300/1	1.25	0..05	2	0.05	0.12	0.05	0.12	0.04	0.04
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1	1

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GRID I/C	MICOM P127	600/1	2	0.1	5	1	0.15	0.13	1.5	1	1

GRID TG-1	MICOM P127	1200/1	0.8	0.14	3	0.3	0.1	0.12	0.1	0.15	0.15
11KV GRID I/C	MICOM P127	1200/1	0.95	0.15	3.5	0.5	1	0.14	1	0.35	0.35
66 KV RELAY SF6	MICOM P127	200/1	1	0.3	6	0.1	0.5	0.15	1	0.2	0.2

GENERATOR -2

Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
GENERATOR	SIPROTEC 7UM62	1000/1	0.92	0.8	1.31	0.1	0.33	0.2	~	~	~
TR1	SPAJ 140 C	600/1	0.54	0.13	4	0.04	0.15	0.12	0.17	0.1	0.1
TR2	SPAJ 140 C	600/1	0.5	0.13	4	0.04	0.15	0.12	0.17	0.1	0.1
O/G Feeder 12	MICOM P111	600/1	0.7	0.05	1.2	0.05	0.1	0.02	0.12	0.02	0.02
O/G Feeder 11	MICOM P122	600/1	0.4	0.04	0.5	0.05	0.05	0.025	0.05	0.02	0.02
O/G Feeder 10	MICOM P111	600/1	0.5	0.05	0.5	0.04	0.1	0.05	0.1	0.05	0.05
O/G Feeder 9	MICOM P122	800/1	1	0.04	1.8	0.04	0.1	0.025	0.05	0.02	0.02
Power Outlet	MICOM P127	1200/1	0.9	0.07	1.8	0.08	0.06	0.04	0.1	0.05	0.05

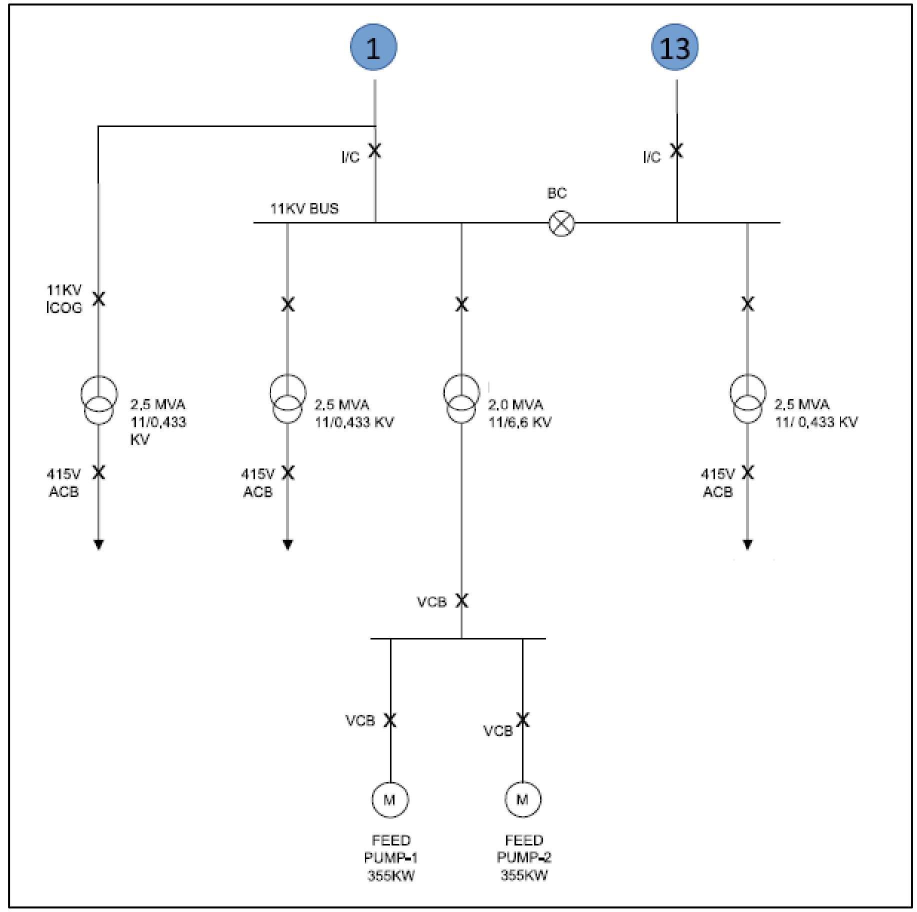
Feeder Name	Relay Type	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
11KV TRANSFORMER	MICOM P122	1200/1	0.9	0.07	1.7	0.07	0.1	0.05	0.1	0.05	0.05
66 KV SF-6 Breaker	MICOM P122	200/1	0.9	0.05	1.6	0.08	0.1	0.07	0.1	0.07	0.07
66 KV SF-6 Breaker	MICOM P122	200/1	0.9	0.05	1.6	0.08	0.1	0.07	0.1	0.07	0.07

Sub- System Structure:

On the 11 kV plant bus, two identical recovery units are fed via dedicated transformers and 415 ACBs:

1. Subsystem-1: Power is stepped down by 2.5 MVA, 11 kV/ 0.433 kV transformer. Its low-voltage side is protected by a 415 V ACB, isolating the downstream 415 V bus. From that bus, two 355 kW feed pumps (marked M) are each feed through vacuum circuit breakers (VCBs). These ensure selective pump isolation and protection against motor faults.

2. Subsystem –2: Supplied by two parallel transformers: one 2.5 MVA, 11 kV/0.433 kV unit (REC-2 TR-1) and one 2 MVA, 11 kV/0.433 kV unit (REC-2 TR-2). Both step-down transformers deliver power to a common 415 kV ACB bus. Like Recovery-1, this LV bus then feeds its own pair of 355 kW vacuum-breaker protected feed pumps.



Key Points –

1. Redundancy and Capacity: Having two transformers in the subsystem provides either increased or increased capacity (both online) or redundancy (one on standby).

2. Selective Isolation: Each transformer's 415 V ACB and each pump's VCB allow faults to be cleared locally without affecting the other subsystem unit.

3.Protection Coordination: Overcurrent and earth fault relays on the 11 kV side coordinate with the LV breakers to ensure only the faulty feeder or pump is tripped.

Trip Time Calculation of Overcurrent Using IEC 60255 Standard Inverse Curve:

Feeder Name	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
Substation I/C	300/1	1.25	0..05	2	0.05	0.12	0.05	0.12	0.04	0.132
Feeder 1	600/1	1.0	0.025	1.7	0.05	0.12	0.025	0.1	0.03	0.081
Feeder 2	300/1	0.4	0.04	0.9	0.02	0.04	0.04	0.07	0.02	0.072

Feeder 3	300/1	0.4	0.04	0.9	0.02	0.04	0.04	0.07	0.02	0.072
Feeder 4	300/1	0.4	0.04	1.0	0.02	0.04	0.04	0.07	0.02	0.072
Feeder 5	300/1	0.9	0.04	1.0	0.02	0.04	0.04	0.07	0.02	0.093

Feeder Name	CT ratio	I>	Time I>(TMS)	I>>	Time I>>	Ie >	Time Ie> (TMS)	Ie>>	Time Ie>> (TMS)	Trip Time
I/C from feeder 13	600/1	0.9	0.07	1.8	0.08	0.06	0.04	0.1	0.05	0.215
Feeder 6	300/1	0.5	0.04	1.1	0.02	0.04	0.04	0.07	0.02	0.077

Selection Criteria of Relays:

- **From Various Manufacturers:** Siemens, Schneider, ABB, L&T, etc.
- **Relay Specifications:** You Selection of appropriate relay for certain type of feeder is based on:
    - Protection Requirements

a. Different types of faults, such as short circuits, overloads, ground faults, and broken conductors, require specific relay types.
b. Overcurrent protection relays are simple and inexpensive, making them suitable for radial feeders. Directional protection relays are useful for looped or parallel feeders, as they determine the direction of current flow.

- Feeder Type and Length:
  c. For short feeders or busbars, differential protection relays are often used due to their fast, selective, and sensitive nature.
  d. For longer or more complex feeders, distance protection relays are preferred as they measure impedance to detect faults and provide primary and backup protection with various time delays.
- Relay Characteristics:
  e. Distance protection relays have multiple zones (Zone 1, Zone 2, Zone 3) for primary and backup protection. Zone 1 covers 80% to 90% of the feeder length with no time delay.
  f. Zone 2 and Zone 3 provide backup protection with increasing time delays.
- Coordination with Other relays:
  g. The selection of relays must ensure proper coordination with other relays to avoid unnecessary tripping and ensure selective protection.
  h. This involves setting appropriate time delays and impedance threshold.
- Cost and Maintenance:
  i. Numerical relays are generally preferred over electromechanical or static relays for better performance, functionality, and diagnostics.
  j. However, the cost and maintenance requirements of relays must also be considered.
- Environment and Safety Considerations:
  k. Arc-flash detection relays are recommended for faster tripping and improved safety for personnel.
  l. Additionally, the relay's ability to handle inrush currents and the rated voltage must be evaluated to ensure reliability and durability.

Basic Overview:

Relay	Manufacturer	Type	Primary Use
Micom P111	Schneider Electric	Overcurrent Protection Relay	Simple radial feeders, backup protection
Micom p122	Schneider Electric	Directional Overcurrent Relay	Directional feeder protection (ring/mesh)
Micom P127	Schneider Electric	Advanced Feeder Protection Relay	Feeder, motor, Transformer with more logic

SIPROTEC 7UM62	Siemens	Motor Protection Relay	Comprehensive motor protection with automation and motoring

Key Differences:

Feature	MICOM P111	MICOM P121	MICOM P127	SIPROTECH 7UM62
Protection Functions	3-stage Overcurrent (50/51)	3-stage + Directional (67)	3-stage + Directional + Earth Fault + Voltage	Full motor protection (thermal, locked rotor, unbalance, stalling etc.)
Directional Capability	Not Present	Present	Present	Present
Voltage Input	Not Present	Present (for direction)	Present	Present
Communication	Modbus, IEC60870-5-103 (basic)	IEC103, Modbus	IEC61850, DNP3 (optional)	IEC61850, PROFIBUS, Modbus, Ethernet
Programmable Logic	Minimal	Moderate	Advanced (logic equations, interlocks)	Advanced automation logic
Display	Basic LCD	Improved	Graphical LCD	High-resolution display with diagnostics
SCADA Integration	Limited	Moderate	Good	Excellent (full SCADA & HMI support)
Application Focus	Simple feeders	Ring or parallel feeders	Complex feeders, motors, transformers	Large/critical motors in industrial plants

Feeder Types That Require Directional Overcurrent Protection

1. Ring Main Systems

- **Description**: Power flows in a loop, and faults can occur in either direction.
- **Need for Directional Protection**: To ensure only the relay closest to the fault trips, avoiding unnecessary outages.

2. Parallel Feeders

- **Description**: Two or more feeders supply the same load.
- **Need for Directional Protection**: To distinguish between internal and external faults and coordinate tripping.

3. Interconnected Networks

- **Description**: Multiple sources feed a common bus or load.
- **Need for Directional Protection**: Fault current can flow from multiple directions; directional relays help isolate the faulted section.

4. Feeder Connected to Generators

- **Description**: Feeders connected to both grid and local generation.
- **Need for Directional Protection**: To prevent back-feed faults from generators and ensure proper fault isolation.

5. Feeder with Reverse Power Flow Possibility

- **Description**: Common in renewable energy systems (solar, wind).
- **Need for Directional Protection**: To detect faults when power flows from load to source.

Source Breakers- Additional Protection Criteria:

Applies mainly to generator incomers (GEN) or critical source feeders.

a) **Directional Protection Relay**
   - Protection against **reverse power flow** (especially important in generator connections).
   - Useful when **multiple sources are interconnected**, preventing back-feed issues.

b) **REF (Restricted Earth Fault) Relay**
   - Sensitive protection that covers **internal phase-to-ground faults** within a zone (like a generator or transformer).
   - Works on the **residual current principle** and is faster than standard earth fault protection.

Model Series	Manufacturer	Key Features	Price Range (INR)
MICOM P12x	Schneider Electric	Full ANSI protection, LCD, Modbus	₹25k–₹60k
MICOM P11x	Schneider Electric	Basic protection, LED, compact	₹15k–₹30k
REYROLLE 7SR	Siemens	IEC 61850, directional, event logs	₹35k–₹60k
REF615	ABB	Graphical UI, GOOSE, feeder protection	₹55k–₹90k
IRIPRO-V4	C&S Electric	USB, RS-485, IDMT	₹18k–₹30k

Low Voltage Relay Settings and Trip Details:

1. **Overload Protection Settings (OL/ Long-Time):**

- **OL Char. I2t/I4t:** Type of overload characteristic curve:
    a**. I2T =** Inverse-time tripping
    b**. I4T** = definite time tripping
- **OC (L) IR**: Long-time pickup setting (in multiples of In) for overload protection.
- **OC(L) Time TR(s)**: Long-time delay settings, defines how long the breaker waits before tripping on overload. Given in seconds.

2. **Short Circuit Protection Settings (OC(S)):**
    - **OC(S) Isd:** Short-time pickup settings, usually in multiples of In. Determines the point where short circuit protection kicks in.
    - **OC(S) Time tsd/I2sd**: Short-time delay settings.
        a. 0.1(I2sd) means inverse delay of 0.1s.
        b. 0.2 (I2sd) means slightly longer delay.
    - **OC(Ii) Time TR(s)**: Instantaneous trip setting – breaker trips immediately at this multiple of In.
3. **Earth Fault Protection Settings (EF(G)):**
    - **EF(G):** Whether earth fault protection is enabled or not this is indicated by this, in which B, C, D and other alphabets indicate the EF protection group.
    - **EF(G) Time tg/I2tg**: Time delay for ground fault tripping.
4. **Trip Time:**
    - **Trip Time:** The calculated or observed total trip time of the breaker in seconds for a fault condition, considering all delays and settings.

LSIG Relay:

Used for PPC incomer and outgoing breakers.

- **L** = Long-time (Overload protection)
- **S** = Short time (Short circuit with delay)
- **I** = Instantaneous (High current faults)
- **G** = Ground Fault (Earth fault protection)

LSIG relays are standard for **ACBs** at **PCC incomers and critical feeders**.

Application:

- **PCC I/C** (Power Control Center Incomer): Main LV incomer from transformer or bus.
- **O/G Breakers** (Outgoing Breakers): Based on **cable distance**, settings vary to ensure relay coordination.

Trip Time Calculation for Overcurrent:

$$Trip\ Time = \frac{(6 \times Isetting)^2 \times Tsetting}{(Fault\ Current)^2}$$

Low Voltage Readings (PCC1, Cogen-2)

Feeder Name	ACB Maker	Trip Unit Model	Rating (On Trip Unit) In	OL Char. I2t/I4t	OC(L) IR	OC(L) Time TR(s)	OC(S) Isd	OC(S) Time tsd(s)/I2tsd	OC(I) Ii	EF(G)	EF(G) Time tg/I2tg	Trip Time (s)
`Main ACB	Siemens	ETU45B	4000	I2t	0.7	2	6	0.1 (I2tsd)	8	B	0.1 (tg)	22.5792
Bus Coupler	Siemens	ETU45B	4000	I2t	0.7	3.5	8	0.4 (I2tsd)	4	C	0.3 (tg)	39.5136
MCC-1	Siemens	ETU45B	1250	I2t	0.8	25	8	0.3 (I2tsd)	3	B	0.1 (tg)	36
MCC-6	Siemens	ETU45B	1250	I2t	1.0	5.5	6	0.4 (I2tsd)	3	C	0.2 (tg)	12.375
VFD Supply	Siemens	ETU45B	1250	I2t	0.9	3.5	6	0.4 (I2tsd)	3	B	0.2 (tg)	6.378
MCC-5A	Siemens	ETU45B	1250	I2t	0.5	10	4	0.4 (I2tsd)	3	C	0.4 (tg)	5.625
Capacitor Feeder 1	Siemens	ETU45B	1250	I2t	0.7	3.5	2.5	0.2 (I2tsd)	2.2	B	0.2 (tg)	3.85
MCC-2	Siemens	ETU45B	1250	I2t	0.4	10	6	0.3 (I2tsd)	3	B	0.1 (tg)	3.6
MCC-4	Siemens	ETU45B	1250	I2t	0.55	3.5	2.5	0.2 (I2tsd)	6	B	0.1 (tg)	2.382
Capacitor Feeder 2	Siemens	ETU45B	1250	I2t	0.55	3.5	2.5	0.2 (tsd)	2.2	B	0.1 (tg)	2.38
MCC-5B	Siemens	ETU45B	1250	I2t	0.45	3.5	1.5	0.2 (I2tsd)	3	B	0.2 (tg)	1.594

Low Voltage Readings (PCC2, Cogen-3)

Feeder Name	ACB Maker	Trip Unit Model	Rating (On Trip Unit) In	OL Char. I2t/I4t	OC(L) IR	OC(L) Time TR(s)	OC(S) Isd	OC(S) Time tsd(s)/I2tsd	OC(I) Ii	EF(G)	EF(G) Time tg/I2tg	Trip Time (s)
I/C From 2.5 MVA T/F	Merlin Gerin	Micrologic6.0	4000	I2t	0.9	8	3	0.3 (I2tsd)	4	B	0.2 (I2tg)	149.2992

MCC-1	Scheider	Micrologic6.0	1250	I2t	1	8	2.5	0.1 (I2tsd)	3	B	0.1 (tg)	18
Cap Bank Feeder	Merlin Gerin	Micrologic6.0	1250	I2t	1	8	3	0.2 (I2tsd)	3	B	0.1 (tg)	18
MCC-2	Merlin Gerin	Micrologic6.0	1250	I2t	0.95	8	2.5	0.2 (I2tsd)	3	A	0.3 (tg)	16.245
MCC-3	Merlin Gerin	Micrologic6.0	1250	I2t	0.8	4	2.5	0.2 (I2tsd)	3	A	032 (I2tg)	5.76
MCC-4	Merlin Gerin	Micrologic6.0	1250	I2t	0.8	4	2.5	0.2 (I2tsd)	3	A	0.3 (tg)	5.76
MCC-5	Scheider	Micrologic6.0	1250	I2t	1	2	4	0.2 (I2tsd)	4	A	0.1 (tg)	4.5
MCC-6	Merlin Gerin	Micrologic6.0	1250	I2t	0.7	1	1.5	0.1 (tsd)	2	A	0.1 (tg)	1.1025
MCC-7	Merlin Gerin	Micrologic6.0	1250	I2t	0.6	0.5	1.5	0.1 (tsd)	2	A	0.2 (tg)	0.405
MCC-8	Merlin Gerin	Micrologic6.0	4000	I2t	0.4	0.5	1.5	0.1 (tsd)	2	A	0.1 (tg)	0.18

Trip Time Calculation for Earth Current:

- A = 100 A
- B = 300 A
- C= 600 A
- D = 900 A
- E = 1200 A

Low Voltage Readings (GEN-2)

Feeder Name	ACB Maker	Trip Unit Model	Rating (On Trip Unit) In	OL Char. I2t/I4t	OC(L) IR	OC(L) Time TR(s)	OC(S) Isd	OC(S) Time tsd(s)/I2tsd	OC(I) Ii	EF(G)	EF(G) Time tg/I2tg	Trip Time (s)
`Main ACB	Siemens	ETU45B	4000	I2t	0.7	2	6	0.1 (I2tsd)	8	B	0.1 (tg)	0.01296
Bus Coupler	Siemens	ETU45B	4000	I2t	0.7	3.5	8	0.4 (I2tsd)	4	C	0.3 (tg)	0.15552
MCC-1	Siemens	ETU45B	1250	I2t	0.8	25	8	0.3 (I2tsd)	3	B	0.1 (tg)	0.01296
MCC-6	Siemens	ETU45B	1250	I2t	1.0	5.5	6	0.4 (I2tsd)	3	C	0.2 (tg)	0.10368
VFD Supply	Siemens	ETU45B	1250	I2t	0.9	3.5	6	0.4 (I2tsd)	3	B	0.2 (tg)	0.02592
MCC-5A	Siemens	ETU45B	1250	I2t	0.5	10	4	0.4 (I2tsd)	3	C	0.4 (tg)	0.20736
Capacitor Feeder 1	Siemens	ETU45B	1250	I2t	0.7	3.5	2.5	0.2 (I2tsd)	2.2	B	0.2 (tg)	0.02592

MCC-2	Siemens	ETU45B	1250	I2t	0.4	10	6	0.3 (I2tsd)	3	B	0.1 (tg)	0.01296
MCC-4	Siemens	ETU45B	1250	I2t	0.55	3.5	2.5	0.2 (I2tsd)	6	B	0.1 (tg)	0.01296
Capacitor Feeder 2	Siemens	ETU45B	1250	I2t	0.55	3.5	2.5	0.2 (tsd)	2.2	B	0.1 (tg)	0.01296
MCC-5B	Siemens	ETU45B	1250	I2t	0.45	3.5	1.5	0.2 (I2tsd)	3	B	0.2 (tg)	0.02592

Low Voltage Readings (GEN-3)

Feeder Name	ACB Maker	Trip Unit Model	Rating (On Trip Unit) In	OL Char. I2t/I4t	OC(L) IR	OC(L) Time TR(s)	OC(S) Isd	OC(S) Time tsd(s)/I2tsd	OC(I) Ii	EF(G)	EF(G) Time tg/I2tg	Trip Time (s)
I/C From 2.5 MVA T/F	Merlin Gerin	Micrologic6.0	4000	I2t	0.9	8	3	0.3 (I2tsd)	4	B	0.2 (I2tg)	0.02592
MCC-1	Scheider	Micrologic6.0	1250	I2t	1	8	2.5	0.1 (I2tsd)	3	B	0.1 (tg)	0.01296
Cap Bank Feeder	Merlin Gerin	Micrologic6.0	1250	I2t	1	8	3	0.2 (I2tsd)	3	B	0.1 (tg)	0.01296
MCC-2	Merlin Gerin	Micrologic6.0	1250	I2t	0.95	8	2.5	0.2 (I2tsd)	3	A	0.3 (tg)	0.00432
MCC-3	Merlin Gerin	Micrologic6.0	1250	I2t	0.8	4	2.5	0.2 (I2tsd)	3	A	0.32 (I2tg)	0.004608
MCC-4	Merlin Gerin	Micrologic6.0	1250	I2t	0.8	4	2.5	0.2 (I2tsd)	3	A	0.3 (tg)	0.00432
MCC-5	Scheider	Micrologic6.0	1250	I2t	1	2	4	0.2 (I2tsd)	4	A	0.1 (tg)	0.00144
MCC-6	Merlin Gerin	Micrologic6.0	1250	I2t	0.7	1	1.5	0.1 (tsd)	2	A	0.1 (tg)	0.00144
MCC-7	Merlin Gerin	Micrologic6.0	1250	I2t	0.6	0.5	1.5	0.1 (tsd)	2	A	0.2 (tg)	0.00288
MCC-8	Merlin Gerin	Micrologic6.0	4000	I2t	0.4	0.5	1.5	0.1 (tsd)	2	A	0.1 (tg)	0.00144

Underground Cable and Overhead cable losses:

Resistivity of Aluminum: $2.82*10^{-8}\ \Omega$ m

Name of the plant	Distance (from respective gen) (meter)	No. Of Cables	Cross Section Area (mm^2)	Current through cable (Amp)	Impedance of cable	Total Power loss (w)
Feeder 1	250	2	300	116.64	0.01175	159.857
Feeder 2	400	2	300	162.9	0.0059	156.565
Feeder 3	120	1	300	52.24	0.0128	34.9314
Feeder 4	250	1	300	52.24	0.0235	64.1319
Feeder 5	400	2	300	92.15	0.0118	100.20
Feeder 6	600	3	400	223.83	0.0141	706.408
Feeder 7	400	2	300	45.48	0.0188	38.8865
Feeder 8	400	2	300	111.96	0.0188	235.6589
Feeder 9	250	1	300	37.87	0.0235	33.702
Feeder 10	150	2	400	26.2432	0.00528	3.6363
Feeder 11	500	3	400	236.189	0.01175	665.4766
Feeder 12	528	3	400	320.750	0.01241	1276.7478

Total Power Loss through underground and overhead transmission line = 3.476 kW

Transformer losses:

- No load = 20kW
- On Load = 105 kW

Transmission line loss:

- Dc resistance at 20° C (per km) = 0.1828 Ω
- Total dc resistance = 2.32613 Ω
- Avg current through transmission line = 48.1125

Total Power loss through transmission line: 5.384 kW

Total Power loss in Transmission and Distribution Network: 3.476 kW + 105 kW + 5.384 kW = 113.86 kW

Other Related Losses and Studies:

These are power quality and distribution efficiency parameters that you may include in your relay study or performance analysis:

1) **Power Factor**
   - Low power factor leads to high current -> overheating -> relay pickup faster than expected.
   - Important for relay settings and capacitor bank control.
2) **THD (Total Harmonic Distortion)**
   - Harmonics affect current sensing and can cause **nuisance tripping**.
   - Involves voltage/current waveform distortion, particularly in VFD/motor loads.
3) **Voltage Drop (LT side)**
   - Voltage drops across long cables can lead to **under-voltage conditions** at the end of the equipment.
   - Affects sizing and sensitivity of protection relays.

The limit for **voltage Total Harmonic Distortion (THD)** is defined by the **IEEE 519-2014 standard,** which is widely used in power quality management. According to this standard:

- For power systems operating at 69 kV and below, the voltage THD limit is **5.0%.**
- Additionally, each individual harmonic should not exceed **3%** of the fundamental voltage

Feeder Name	PF Avg				Voltage THD Avg				Current THD Avg				Power THD Avg			
Feeder 1					2.62	2.77	2.73	2.71	0	0	0	0	0	0	0	0
Feeder 2	0.935	0.938	0.949	0.940	1.81	1.77	1.80	5.38	3.81	3.53	3.75	3.696	0.03	0.02	0.01	0.02
Feeder 3	0.886	0.879	0.889	0.885	1.84	1.85	1.79	1.83	7.89	7.72	8.10	7.903	0.09	0.09	0.09	0.09
Feeder 4	0.926	0.910	0.934	0.924	1.84	1.68	1.81	1.78	15.9	15.9	17.2	16.34	0.03	0.05	0.00	0.026
Feeder 5	0.82	0.804	0.863	0.830	1.97	1.48	1.94	1.80	2.36	2.14	2.44	2.313	0.00	0.00	0.00	0.00
Feeder 6	0.846	0.840	0.842	0.843	2.55	1.33	2.04	1.97	4.00	3.93	4.07	4.000	0.00	0.01	0.00	0.003
Feeder 7	0.941	0.935	0.940	0.939	2.14	1.41	1.94	1.83	7.76	7.59	7.88	7.743	0.01	0.06	0.12	0.063
Feeder 8	0.958	0.959	0.963	0.960	2.27	1.23	2.20	1.9	10.4	10.2	10.3	10.32	0.00	0.02	0.01	0.01
Feeder 9	0.850	0.849	0.859	0.852	1.84	1.77	1.95	1.85	6.64	6.89	7.24	6.923	0.00	0.01	0.04	0.016
Feeder 10	0.947	0.955	0.949	0.951	1.90	1.85	1.86	1.87	6.77	6.98	7.19	6.980	0.00	0.01	0.03	0.013

Feeder 11	0.846	0.833	0.847	0.842	1.32	1.69	2.84	1.95	2.78	2.79	2.91	2.826	0.02	0.01	0.00	0.01
Feeder 12	0.258	0.265	0.192	0.238	1.95	1.41	1.99	1.78	7.76	9.01	8.40	8.390	0.02	0.31	0.52	0.283
Feeder 13	0.967	0.994	0.963	0.978	4.42	3.84	3.85	4.04	8.09	8.78	8.93	8.600	0.04	0.05	0.1	0.063
Feeder 14	0.971	0.996	0.964	0.981	4.43	3.85	3.82	4.03	23.3	23.7	23.7	23.56	0.00	0.00	0.00	0.00
Feeder 15	0.892	0.958	0.891	0.917	4.41	3.86	3.81	4.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Feeder 16	0.933	0.945	0.939	0.939	2.73	2.92	2.91	2.85	2.17	2.18	2.16	2.172	0.19	0.22	0.23	0.213
Feeder 17	0.854	0.850	0.852	0.852	2.74	2.92	2.92	2.86	4.86	4.92	5.12	4.960	0.00	0.01	0.00	0.003
Feeder 18	0.974	0.976	0.973	0.974	2.77	2.96	2.97	2.9	9.67	10.7	10.4	10.36	0.03	0.06	0.07	0.053
Feeder 19	0.868	0.877	0.871	0.872	2.75	2.85	2.87	2.82	5.48	5.65	5.56	5.563	0.00	0.01	0.01	0.006
Feeder 20	0.737	0.750	0.750	0.746	3.01	2.93	3.04	2.99	3.51	3.40	3.63	3.513	0.05	0.03	0.07	0.05
Feeder 21	0.802	0.825	0.776	0.801	2.72	2.66	2.80	2.73	7.20	8.28	7.83	7.770	0.04	0.11	0.03	0.17
Feeder 22	0.798	0.799	0.812	0.803	2.76	2.81	2.77	2.78	3.91	4.02	4.19	4.04	0.02	0.01	0.05	0.026
Feeder 23	~	~	~	0.922	~	~	~	~	~	~	~	~	~	~	~	~
Feeder 24	0.996	0.989	0.989	0.992	2.79	2.82	2.78	2.80	15.60	14.86	15.69	15.38	0.05	0.07	0.08	0.067
Feeder 25	0.760	0.766	0.764	0.763	2.77	2.79	2.80	2.79	10.25	10.26	10.49	10.33	0.22	0.16	0.24	0.206
Feeder 26	0.923	0.924	0.928	0.925	2.71	2.81	2.79	2.77	21.19	20.55	21.40	21.05	0.14	0.17	0.19	0.167

Average Power factor of last 15 Days

Type of feeder	PF
Feeder 1	0.95
Feeder 2	0.91
Feeder 3	0.88
Feeder 4	0.82
Feeder 5	0.83
Feeder 6	0.82
Feeder 7	0.91
Feeder 8	0.90
Feeder 9	0.64
Feeder 10	0.90

Feeder 11	0.84
Feeder 12	0.96
Feeder 13	0.89
Feeder 14	0.80
Feeder 15	0.13
Feeder 16	0.92
Feeder 17	0.84
Feeder 18	0.83
Feeder 19	0.57
Feeder 20	0.90
Feeder 21	0.9
Feeder 22	0.8
Feeder 23	0.99
Feeder 24	0.94
Feeder 25	0.84
Feeder 26	0.93
Feeder 27	1
Feeder 28	0.2
Feeder 29	0.91
Feeder 30	0.87
Feeder 31	0.85
Feeder 32	-0.32
Feeder 33	0.06
Feeder 34	-0.06
Feeder 35	-0.06

<u>The system containing Generator station should have:</u>

- HT pf = 0.95
- LT pf = 0.92
- Grid I/C = 1.00

<u>The system without Generating station:</u>

- HT pf: 1.00

- LT pf: 0.92